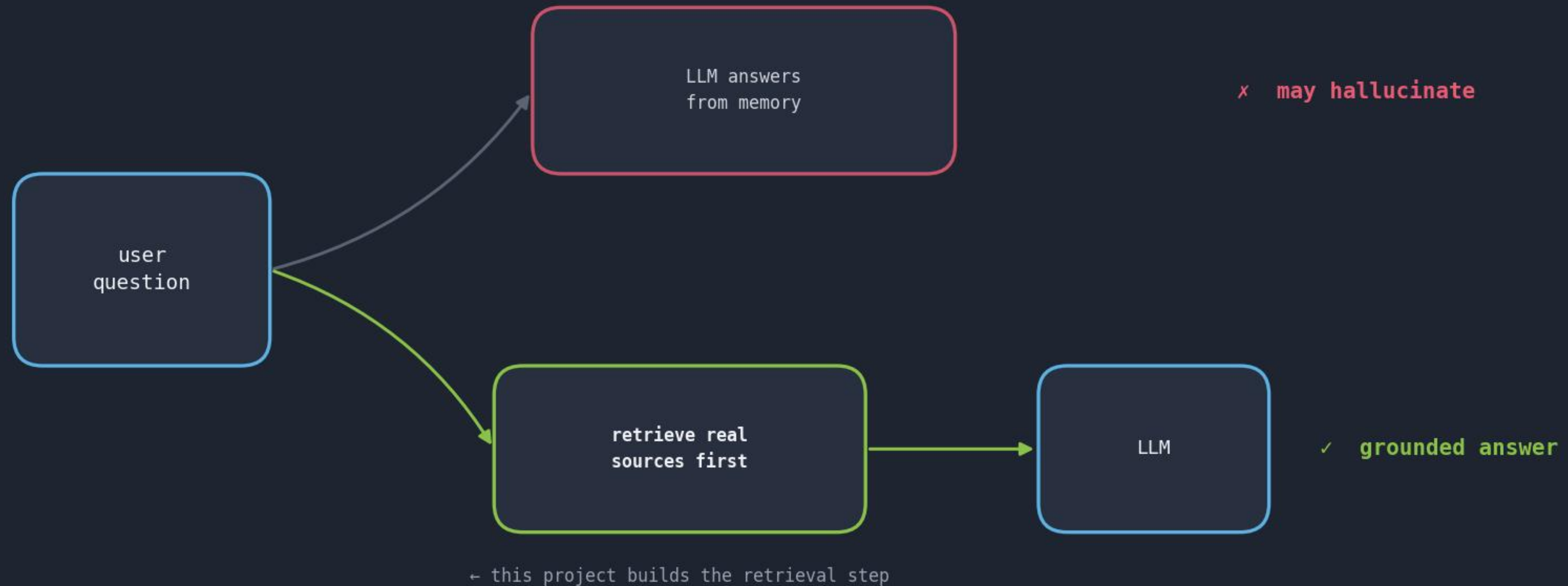

RAG Retrieval Pipeline

Finding the 3 paragraphs that matter — BM25 · trigram · RRF

PRESENTED BY

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01 The Problem



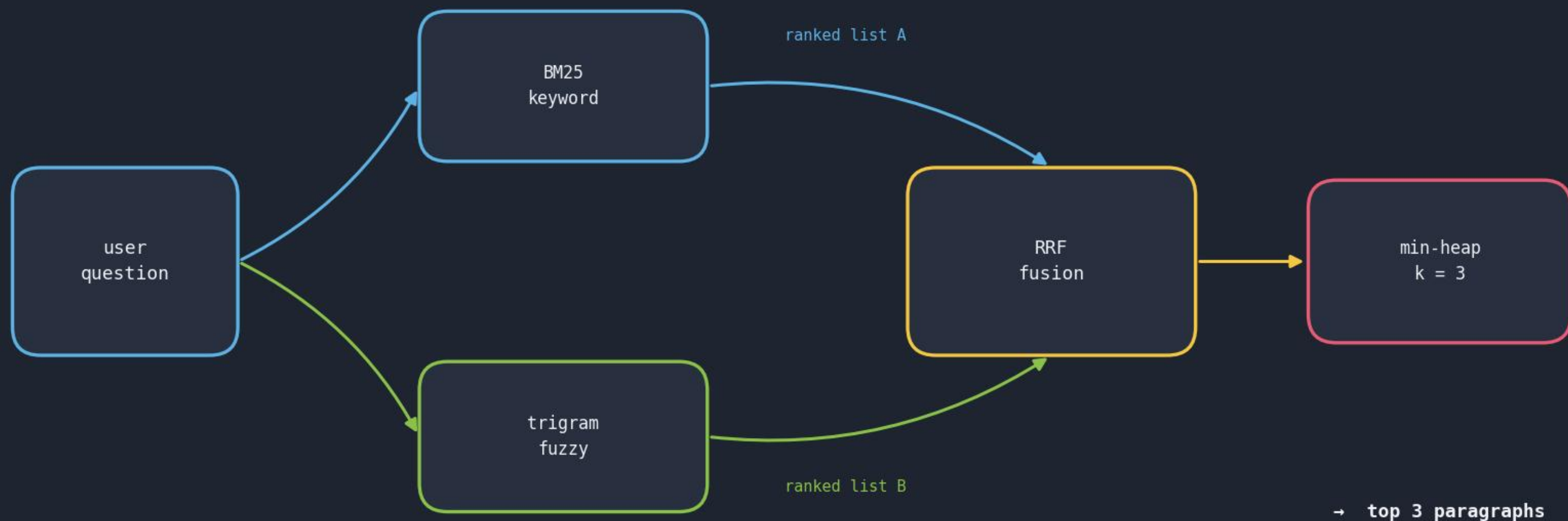
THE GAP

An LLM alone answers from memory — and invents facts.

WHY RAG

Retrieve real paragraphs first, then ground the answer.

02 The Pipeline



THE CHAIN

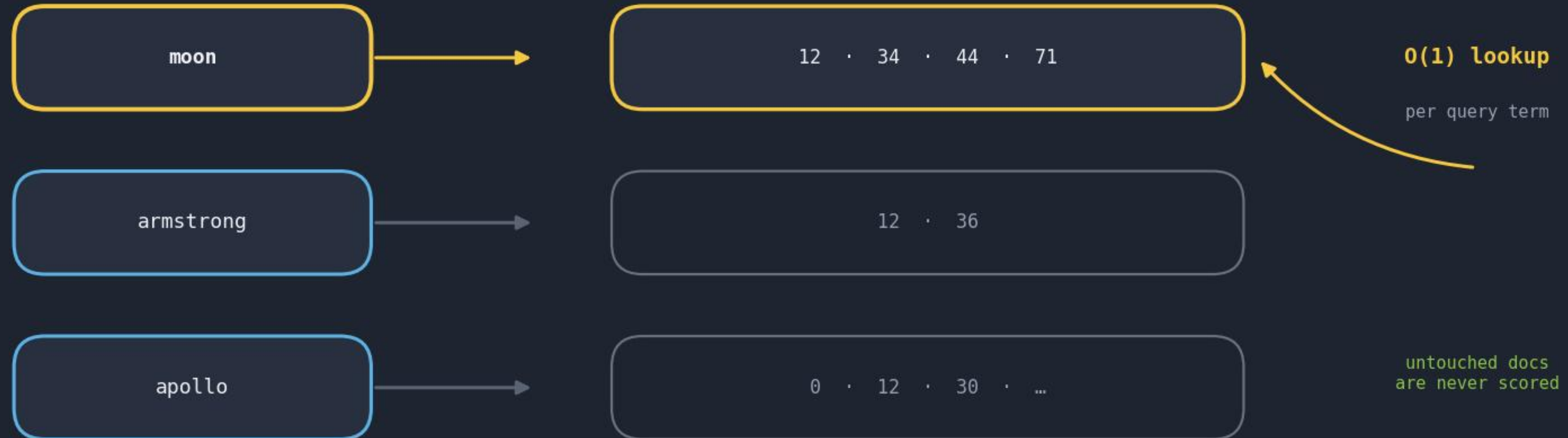
Two rankers search in parallel: exact + fuzzy.

THE OUTPUT

Fuse both, keep the top 3 with a min-heap.

03 BM25 · Index

inverted index : $\text{term} \rightarrow \{ \text{doc_id} : \text{term frequency} \}$



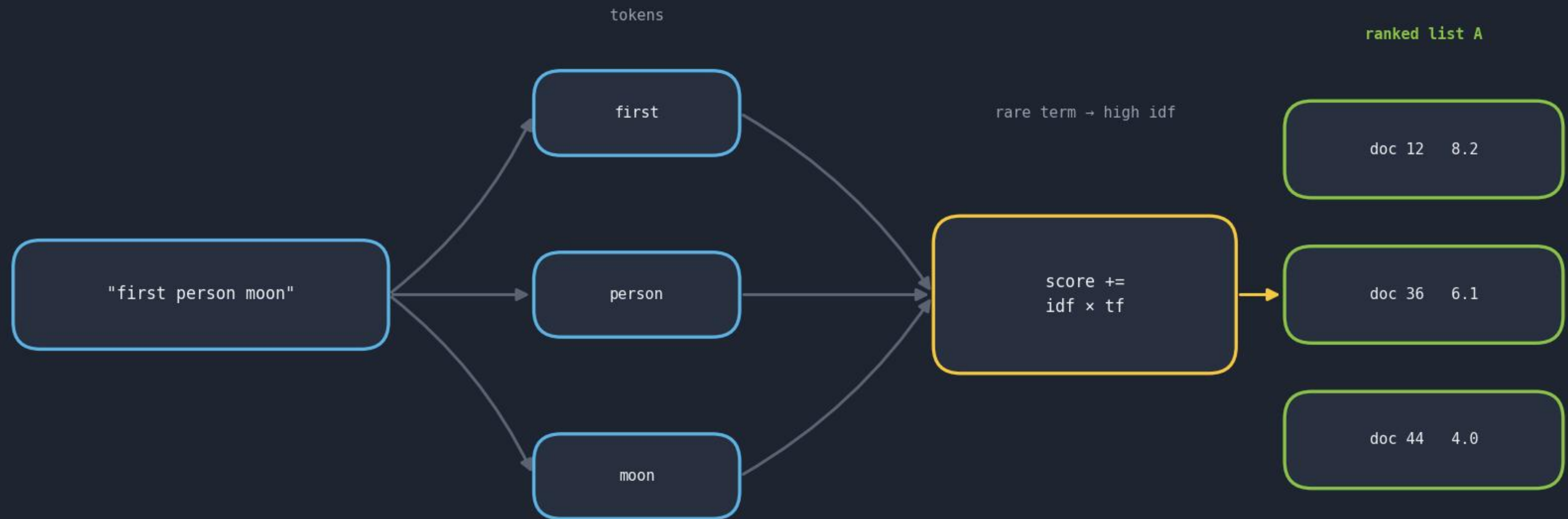
THE INDEX

Each term points to the docs that contain it.

WHY HASH MAP

O(1) lookup; we never scan irrelevant docs.

04 BM25 · Query



THE WALK

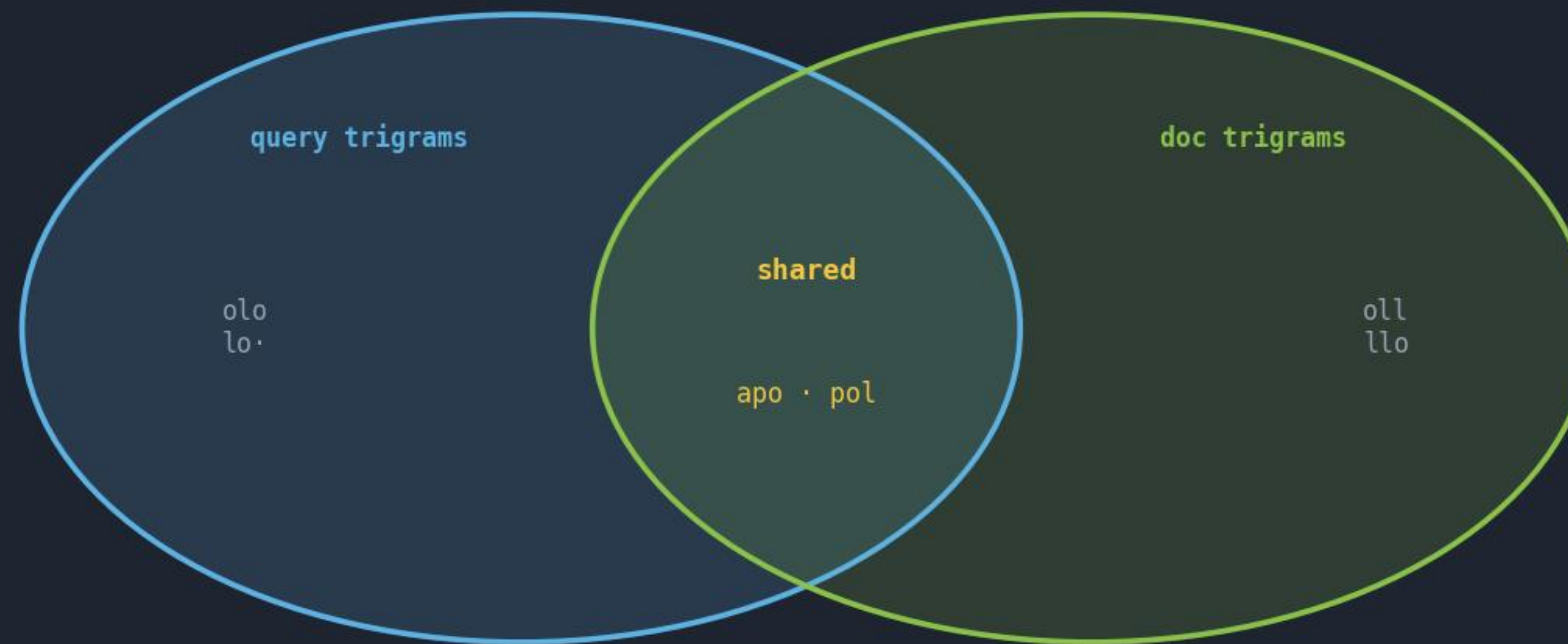
Look up each query term, add $\text{idf} \times \text{tf}$.

THE SCORE

Rare, repeated terms push a doc to the top.

05 Trigrams

"apollo" → { apo · pol · oll · llo }



$$\text{score} = \frac{|q \cap d|}{|q|}$$

THE IDEA

Compare texts as sets of 3-letter chunks.

WHY A SET

Intersection is the hot path — sets make it fast.

06 Typo Tolerance

one missing letter · most trigrams still overlap

typed:

"apolo 11"



real:

"apollo 11"



5 / 6
≈ 0.83
still a
match

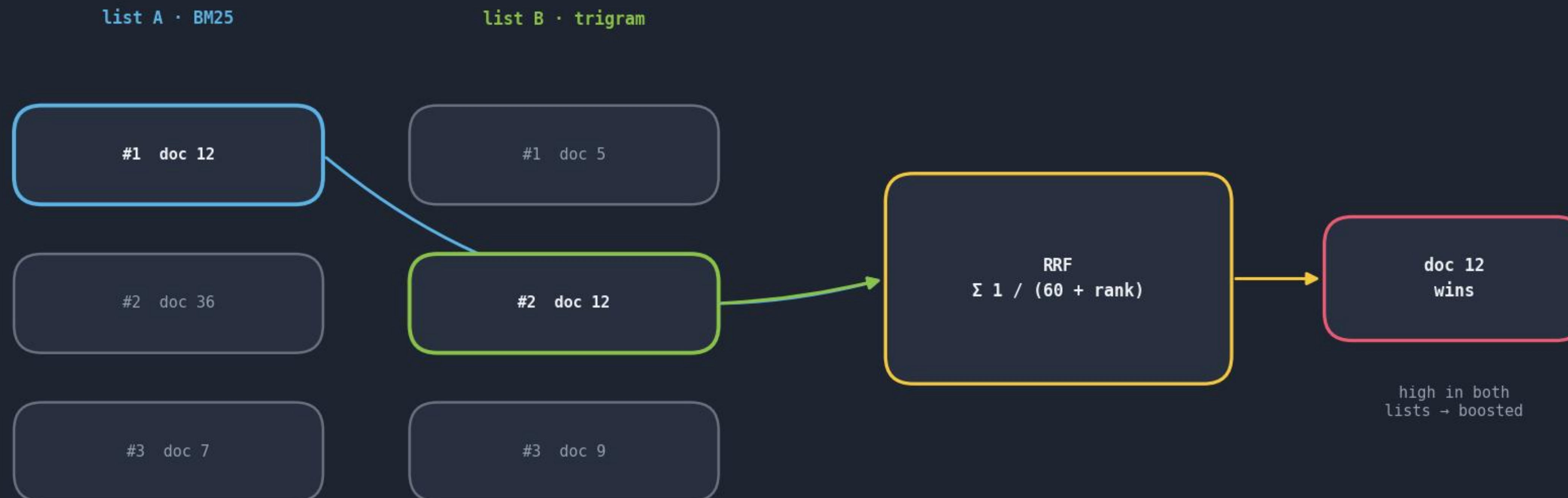
THE EXAMPLE

"apolo" still overlaps "apollo".

WHY FUSE

BM25 misses typos; trigrams catch them.

07 RRF Fusion



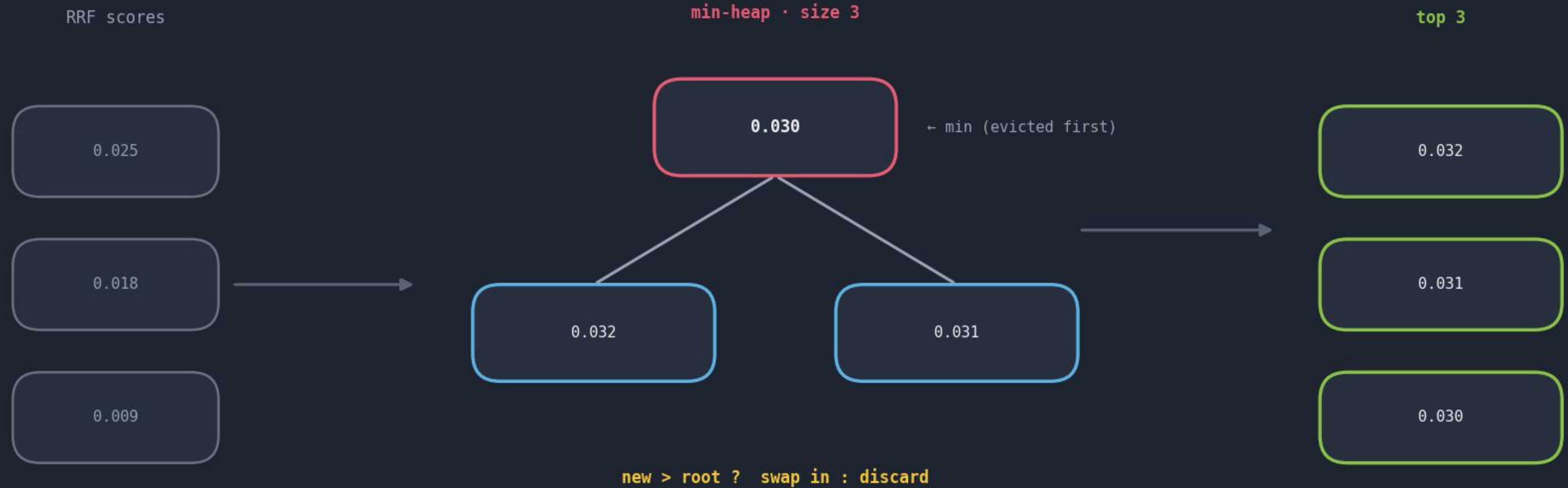
THE MERGE

Each doc scores $1 / (60 + \text{rank})$ per list.

WHY RANKS

Score scales differ — rank position doesn't.

08 Min-Heap Top-3



THE SELECT

Keep only the 3 best scores in a size-3 heap.

WHY A HEAP

$O(D \log k)$ — cheaper than sorting every doc.

09 Complexity

BM25

hash-map index

build	$O(N)$
query	$O(Q \cdot P)$
space	$O(V + N)$

Trigram

set overlap

query	$O(D \cdot L)$
space	$O(L)$

RRF + heap

dict + min-heap

fuse	$O(R \cdot D)$
top-k	$O(D \cdot \log k)$
space	$O(D + k)$

↑ rescans every doc each query — the bottleneck

PER QUERY

BM25 skips docs; trigram rescans all D.

THE BOTTLENECK

Trigram $O(D \cdot L)$ — index it to scale past 10^4 docs.

Thank you